

1.1. Exponential increasing.

$$\varphi: \mathbb{R} \rightarrow \mathbb{R}, \varphi(t) = 2 \cdot e^{3t}, t \in \mathbb{R}.$$

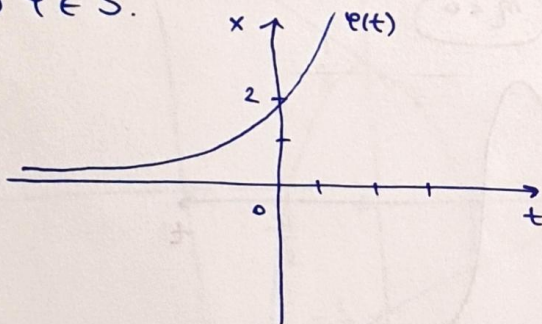
Check if φ is a solution of the IVP: $x' = 3x, x(0) = 2$.

Represent the solution curve and describe its long-term behavior.

$$\varphi'(t) = 2 \cdot 3 \cdot e^{3t} = 6 \cdot e^{3t} = 3 \cdot 2 \cdot e^{3t} = 3 \cdot \varphi(t)$$

$$\varphi(0) = 2 \cdot e^0 = 2 \cdot 1 = 2.$$

$$\Rightarrow \varphi \in S.$$



Real-life ex: money in a bank account (increases exponentially)

Long-term behaviour: $\lim_{t \rightarrow \infty} \varphi(t) = \infty \Rightarrow$ exponentially increasing.
 $\varphi(t) > 0, (\forall) t \in \mathbb{R}.$

1.2. Oscillations with constant amplitude.

$$\varphi: \mathbb{R} \rightarrow \mathbb{R}, \varphi(t) = \frac{1}{6} \sin t, t \in \mathbb{R}.$$

$$\text{Sol. of IVP: } x'' + x = 0, x(0) = 0, x'(0) = \frac{1}{6} \quad ?$$

Repr. solution curve and describe its long-term behavior.

Motion of the simple planar gravity pendulum.

$$\varphi(t) = \eta \cdot \sin t \rightarrow \text{distance between vertical and current position.}$$

$$x' = \varphi'(t) = \eta \cdot \cos t$$

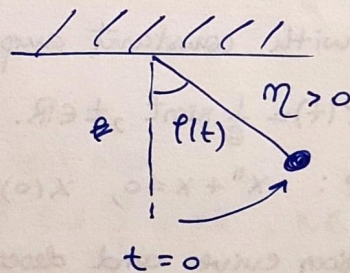
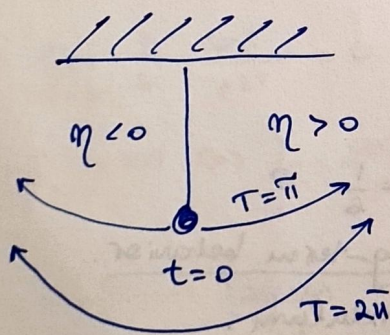
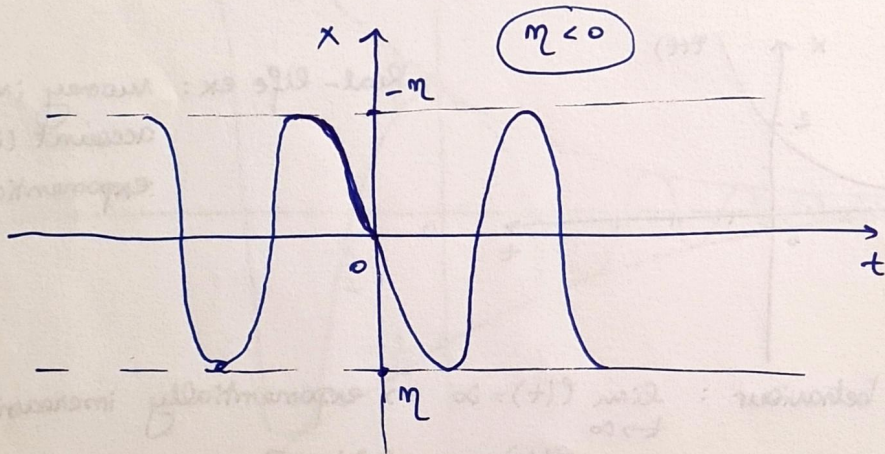
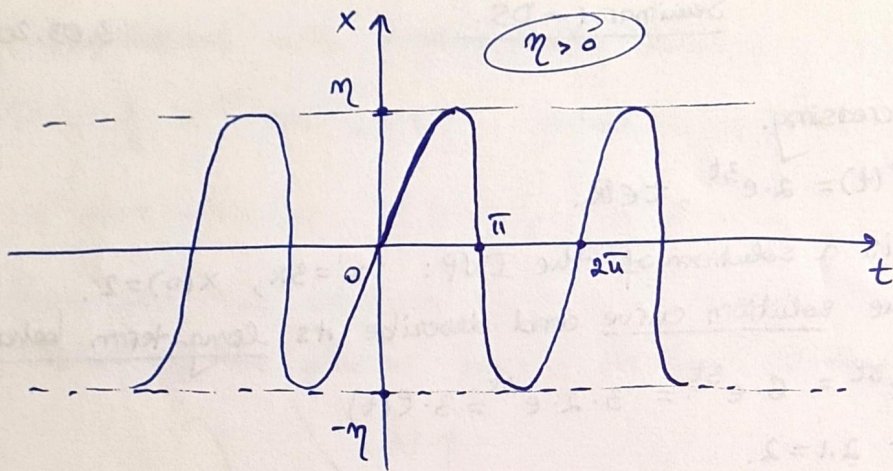
$$\varphi''(t) = -\eta \cdot \sin t$$

$$x'' + x = 0 \Leftrightarrow -\eta \cdot \sin t + \eta \cdot \sin t = 0. \quad \checkmark$$

$$x(0) = \varphi(0) = \eta \cdot \sin 0 = 0 \quad \checkmark$$

$$x'(0) = \varphi'(0) = \eta \cdot \cos 0 = \eta \cdot 1 = \eta \quad \checkmark$$

$$\Rightarrow \varphi \in S.$$



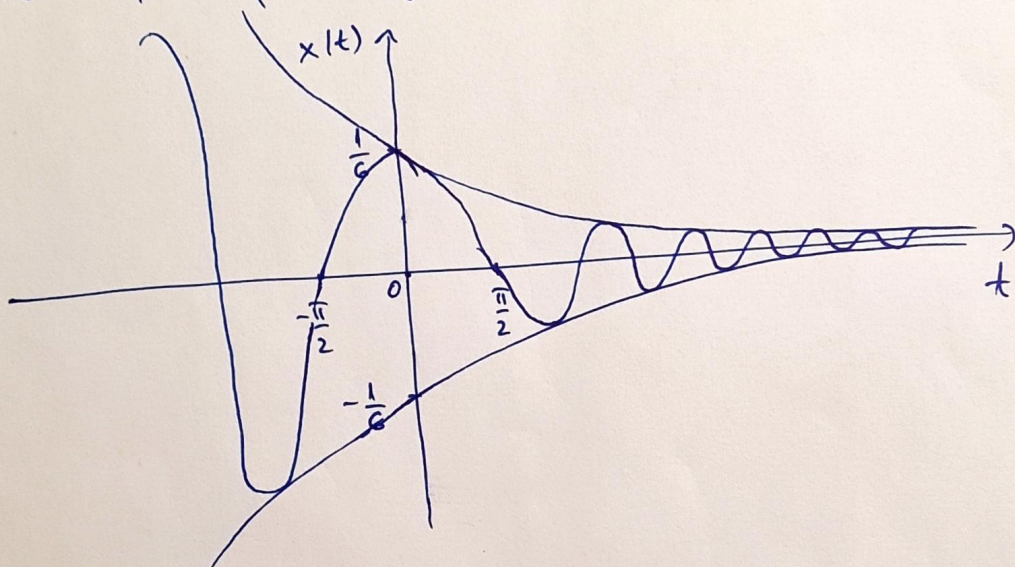
- Long-term behaviour:
- bounded by $-\eta, \eta \rightarrow [-\eta, \eta]$.
 - periodical function with main period $= 2\pi$.
 - oscillates around 0 with constant amplitude
 $\hookrightarrow |\eta|$.

1.3. Oscillations with exponentially decreasing amplitude.

$$f(t) = \frac{1}{6} \cdot e^{-2t} \cdot \cos t, \quad t \in \mathbb{R}. \quad \text{solution of IVP:}$$

$$x'' + 4x' + 5x = 0, \quad x(0) = \frac{1}{6}, \quad x'(0) = -\frac{1}{3}.$$

- Represent this solution curve and describe its long-term behaviour.
- Motion of the planar gravity damped pendulum.



$$\lim_{t \rightarrow \infty} \frac{1}{6} \cdot \frac{\cos t}{e^{2t}} = 0.$$

$$f(t) = 0 \Leftrightarrow \frac{1}{6} \cdot \frac{\cos t}{e^{2t}} = 0 \Leftrightarrow \cos t = 0 \Rightarrow t = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}.$$

$$\cos(t) = 1 \mid \arccos$$

$$\arccos(\cos(t)) = \arccos 1$$

$$\cos t = 2k\pi.$$

Long-term behaviour: - oscillates around 0 with exponentially decreasing amplitude $\lim_{t \rightarrow \infty} \frac{1}{6} \cdot \frac{\cos t}{e^{2t}} = 0.$